



Tutorial: Starter Guide

By the end of this tutorial, you should be able to do the following in the DataHive Databricks workspace:

- Access data from a data request;
- Upload data in `csv` or excel (`xls`, `xlsx`);
- Do basic data wrangling in Spark; and
- Save results and download from DataHive portal

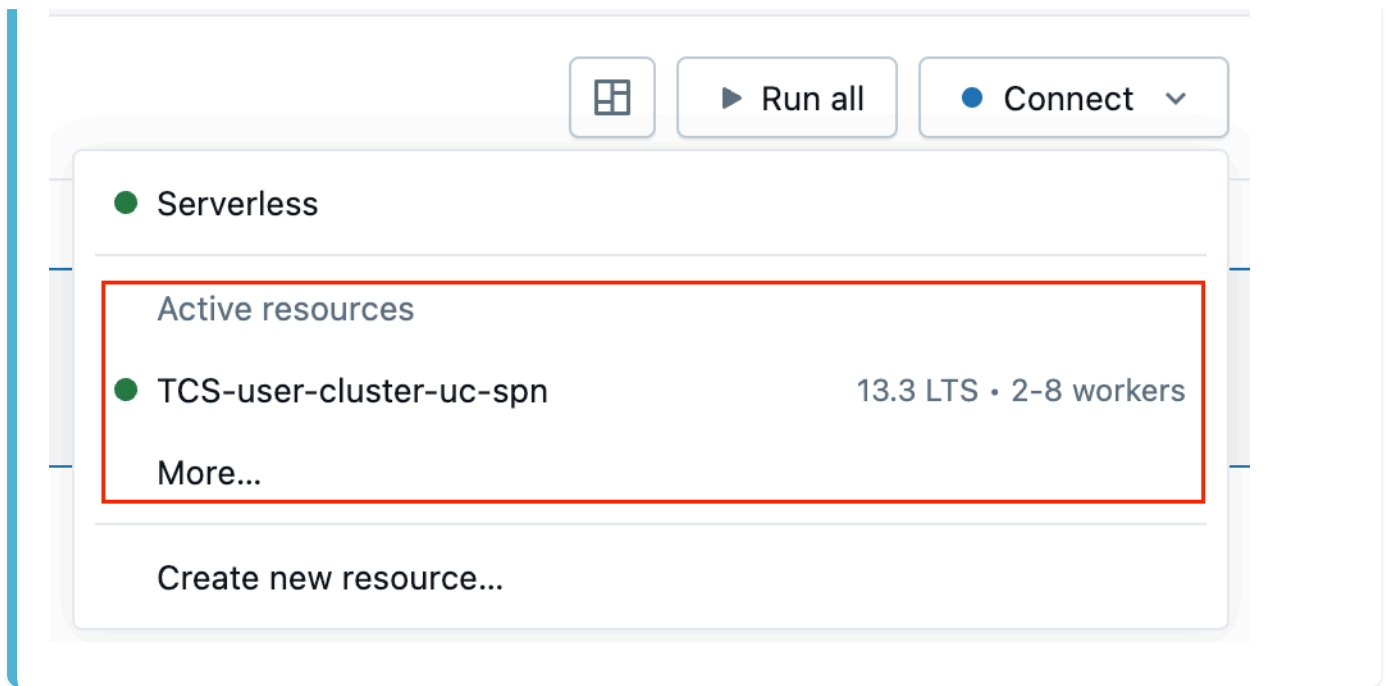
The finished notebook for this tutorial is also provided [here](#) for your reference.

1. Create a notebook

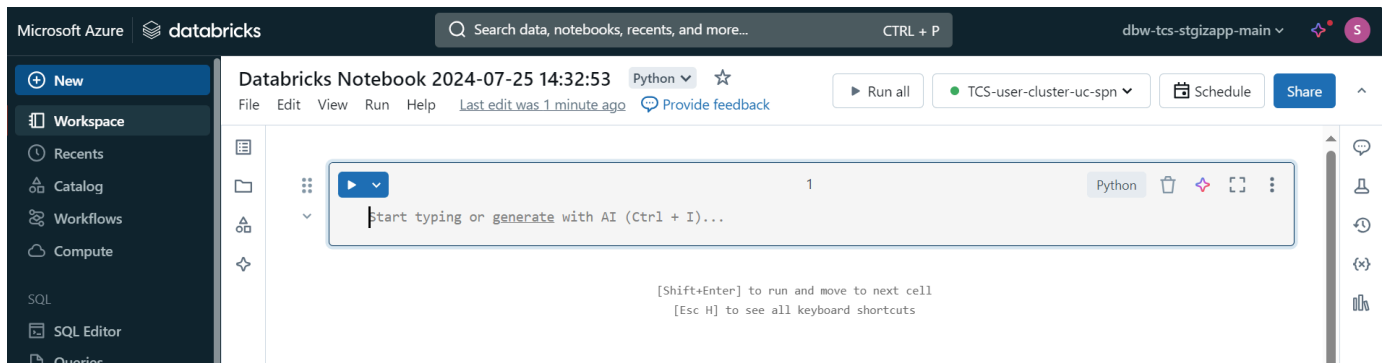
1. Open the left panel, click on `Workspace`.
2. Right click on the folder explorer > Click "Create" and select "Notebook".
3. Provide an appropriate name and select `Python` or `SQL` as the default language. For `Cluster` select `TCS-user-cluster-uc-spn`.

INFO

Note that Serverless cluster is **not supported**. Please ensure you select **TCS-user-cluster-uc-spn** cluster. If this cluster is not listed, then click on the "More" option as shown below.



You should see an empty notebook.



2. Access data from Data Request (DR)

When a request with databricks access is fulfilled, all users in the request will be granted access to table(s) containing the requested data. The tables follow this format: `<request ID>.<table name>`.

To find your `request ID` or `table name`, open the left panel and click `Data`. In the screenshot below, the requested table can be retrieved at `govtech_2021_12_010.ema_towngas`.

Microsoft Azure | databricks

Search data, notebooks, recents, and more...

New

Workspace

Recents

Catalog

Workflows

Compute

SQL

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Queries

Dashboards

Alerts

Catalog Explorer

Send feedback

Catalog

Refresh

^

Type to filter

Filter

>

>

>

govtech_tcs

>

govtech_2021_12_010

>

ema_towngas

>

Follow the syntax below to load your table as a Spark DataFrame.

```
import pyspark.sql.functions as F

database_name = "govtech_2021_12_010"
table_name = "ema_towngas"

df = spark.sql(f"select * from {database_name}.{table_name}")
df.display()
```

3. Uploading data onto Databricks

Data can be uploaded to Databricks via the DataHive Platform. Go to the specific DR page to access the upload interface.

< Data requests / GOVTECH-2022-10-017

Withdraw request

Date requested:
19 Oct 2022, 07:58 PM

Owner:
koh_jun_wei@tech.gov.sg

Team details:
[View](#)

Data request details:
[View](#)

Expiry date:
19 Sep 2027

Status: [View history](#)
● Team Access Changes

Datasets requested

Datasets Download

Team access 1

Databricks ▼

Upload to Databricks

Sync from Databricks

Go to Databricks

Upload table - (max 2.0 GB, supported formats csv, xls, xlsx)

☐ Acknowledge that my uploaded file is not above the classification of Confidential (Cloud Eligible) or Sensitive (normal).

Upload table

Please acknowledge the above agreement before uploading the file

Table name ▲	Uploaded date ▲	Status	
postal_code_mappings	08 Jul 2021, 12:19 PM	● Table created	Download Delete
result123123	08 Jul 2021, 12:19 PM	● Pending anti-virus scan	Download Delete

CAUTION

Files uploaded cannot be above the classification of Confidential (Cloud Eligible) and Sensitive (Normal). There is a file size limit of 2 GB.

Check the acknowledgement and click on the upload box to upload the file. The file formats currently supported are CSV and Excel (xls & xlsx). The first row will be read as a header row. For Excel files, the table will be created only based on your first worksheet.

Datasets requested

Datasets Download

Team access 1

Databricks ▼

Upload to Databricks

Sync from Databricks

Go to Databricks

Upload table - (max 2.0 GB, supported formats csv, xls, xlsx)

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Upload table

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postal_code_mappings	08 Jul 2021, 12:19 PM	● Table created	Download Delete
result123123	08 Jul 2021, 12:19 PM	● Pending anti-virus scan	Download Delete

Before uploading, a preview of the Databricks table to be created will be shown. If this is as expected, you can click on the “Create table” button. You can also choose to specify a name for the Databricks table here.

Confirm Databricks table creation

Databricks table name

file_example_xlsx_5000

Databricks table preview

	first_name	last_name	gender	country	age	date	id
1	Dulce	Abril	Female	United States	32	15/10/2017	1562
2	Mara	Hashimoto	Female	Great Britain	25	16/08/2016	1582
3	Philip	Gent	Male	France	36	21/05/2015	2587
4	Kathleen	Hanner	Female	United States	25	15/10/2017	3549
5	Nereida	Magwood	Female	United States	58	16/08/2016	2468
6	Gaston	Brumm	Male	United States	24	21/05/2015	2554
7	Etta	Hurn	Female	Great Britain	56	15/10/2017	3598
8	Earlean	Melgar	Female	United States	27	16/08/2016	2456
9	Vincenza	Weiland	Female	United States	40	21/05/2015	6548
10	Fallon	Winward	Female	Great Britain	28	16/08/2016	5486

Cancel

Create table

After uploading, the corresponding table can be found on Databricks.

Datasets requested

Datasets Download

Team access 1

Databricks

Upload to Databricks

Sync from Databricks

Go to Databricks

Upload table - (max 2.0 GB, supported formats csv, xls, xlsx)

Acknowledge that my uploaded file is not above the classification of Confidential (Cloud Eligible) or Sensitive (normal).

Upload table

Please acknowledge the above agreement before uploading the file

Table name	Uploaded date	Status
my uploaded	08 Jul 2021, 12:19 PM	Table created

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Search data, notebooks, recents, and more...

New

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Type to filter

govtech_tcs

govtech_2021_12_010

ema_towngas

my_uploaded_csv

By default, the table name will inherit the file name. The table name will be preprocessed such that it conforms to the [snake case](#) format. To avoid this behavior, specify the table name by typing it into the text box beside the upload button.

4. Basic data wrangling with Spark

After creating Spark DataFrames, you can filter and transform the data according to your requirements. Here are some examples:

```
# Rename an existing column
df = df.withColumnRenamed("demo", "number")

# Create new column
df = df.withColumn("double_number", 2 * F.col("number"))

# Filter out rows with given condition
df = df.filter(F.col("double_number") > 5)

# Create new column with given condition
df = df.withColumn(
    "remarks",
```

```
F.when(
    F.col("double_number") > 15,
    F.lit("more than 15")
).otherwise(
    F.col("double_number")
),
)

# Sort/order by certain columns
df.orderBy("number", "double_number").display()
```

Refer to the API reference for Spark 3.1.2 for more information ([Python](#), [SQL](#)).

5. Writing Data using Databricks Notebook

Finally, to extract the processed data out of the platform, run the following command:

```
df.write.saveAsTable("<database_name>.<result_table_name>")
```

CAUTION

Do not save the new table to the same name as an existing table if you do not want to overwrite the existing table.

6. Download Databricks Table from DataHive Platform

To download the result table created in Step 5, go to the "Sync from Databricks" tab:

< Data requests / GOVTECH-2022-10-017

GOVTECH-2022-10-017

Withdraw request

Requestor: Jun Wei KOH (GOVTECH) (koh_jun_wei@tech.gov.sg)

Date requested: 19 Oct 2022, 07:58 PM

Expiry date: N/A

Status: Request Fulfilled [View](#)

Team details: [View](#)

Data request details: [View](#)

Datasets requested

Datasets Download

Databricks ▼

Upload to Databricks

Sync from Databricks

Go to Databricks

Sync table

Databricks table name

Name of Databricks table to sync with

Sync

Table name	Synced on	Status	
postal_code_mappings1	08 Jul 2021, 12:19 PM	Table synced	↺ ⬇ 🗑
postal_code_mappings223123_12311231231	08 Jul 2021, 12:19 PM	Table sync failed 🔔	↺ ⬇ 🗑
result	08 Jul 2021, 12:19 PM	Syncing with Databricks	↺ ⬇ 🗑

Key in the table name you want to sync with in the text field.

After entering the table name, click on the Sync button. After the table has been synced with the DataHive platform, click on the download icon to retrieve your file.

Click on the re-sync icon to re-sync the table to fetch any updates made on Databricks before downloading to get the latest version.

Table name	Synced on	Status	
postal_code_mappings1	08 Jul 2021, 12:19 PM	Table synced	↺ ⬇ 🗑
postal_code_mappings223123_12311231231	08 Jul 2021, 12:19 PM	Table sync failed 🔔	↺ ⬇ 🗑
result	08 Jul 2021, 12:19 PM	Syncing with Databricks	↺ ⬇ 🗑

7. Sharing notebook

To share your notebook with your team members, click the [Share](#) button at the top right. Key in the email address in the pop-up (users must be part of your request) and select the desired permissions to grant.

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P dbw-tcs-stgizapp-main

New

Workspace

Recents

Catalog

Workflows

Compute

SQL

SQL Editor

Queries

Databricks Notebook 2024-07-25 14:32:53 Python

File Edit View Run Help Last edit was 1 minute ago Provide feedback

Run all TCS-user-cluster-uc-spn Schedule Share

1

Python

Start typing or generate with AI (Ctrl + I)...

[Shift+Enter] to run and move to next cell [Esc H] to see all keyboard shortcuts

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🔔

👤

📄

